

ORIGINAL ARTICLE

BCG Revaccination for the Prevention of *Mycobacterium tuberculosis* Infection

A.C. Schmidt,¹ L. Fairlie,² E. Hellström,³ A. Luabeya Kany Kany,⁴ K. Middelkoop,⁵ K. Naidoo,⁶ G. Nair,⁵ A. Gela,⁴ E. Nemes,⁴ T.J. Scriba,⁴ A. Cinar,¹ N. Frahm,¹ R. Mogg,¹ D. Kaufman,¹ M.W. Dunne,¹ and M. Hatherill,⁴ for the BCG REVAX Study Team*

ABSTRACT

BACKGROUND

In a previous phase 2 trial, bacille Calmette–Guérin (BCG) revaccination was not shown to provide protection from primary *Mycobacterium tuberculosis* infection but prevented sustained *M. tuberculosis* infection, defined by an initial conversion on a QuantiFERON-TB (QFT) test (an interferon- γ release assay) from negative to positive, followed by two additional positive QFT tests at 3 and 6 months after the initial conversion (a secondary end point). A vaccine efficacy of 45% (95% confidence interval [CI], 6 to 68) was observed.

METHODS

We performed a phase 2b, double-blind, randomized, placebo-controlled trial to evaluate the efficacy of BCG revaccination, as compared with placebo, for the prevention of sustained QFT test conversion (primary end point) in QFT test–negative, human immunodeficiency virus (HIV)–negative adolescents. Adverse events were assessed in a secondary analysis, and immunogenicity was assessed in an exploratory analysis. Vaccine efficacy was evaluated in the modified intention-to-treat population, which included all the participants who had undergone randomization, received the BCG vaccine or placebo, and had a negative QFT test 10 weeks after receipt of BCG vaccine or placebo; the last criterion was added to exclude participants with *M. tuberculosis* infection around the time that the vaccine or placebo was administered. Hazard ratios and 95% confidence intervals were estimated from a stratified Cox proportional-hazards model.

RESULTS

A total of 1836 participants underwent randomization; 918 received the BCG vaccine, and 917 received placebo. After a median 30 months of follow-up, a sustained QFT test conversion was observed in 62 of 871 participants in the BCG-vaccine group and 59 of 849 participants in the placebo group. The hazard ratio for a sustained QFT test conversion (BCG vaccine vs. placebo) was 1.04 (95% CI, 0.73 to 1.48), for a vaccine efficacy point estimate of –3.8% (95% CI, –48.3 to 27.4). Adverse events occurred more frequently in the BCG-vaccine group than in the placebo group, and most were due to injection-site reactions (pain, redness, swelling, and ulceration). BCG revaccination induced cytokine-positive type 1 helper CD4 T cells.

CONCLUSIONS

BCG revaccination in QFT–test negative, HIV-negative adolescents did not provide protection from sustained *M. tuberculosis* infection. (Funded by the Gates Foundation; ClinicalTrials.gov number NCT04152161.)

The authors' full names, academic degrees, and affiliations are listed at the end of the article. Dr. Schmidt can be contacted at alexander.schmidt@gatesmri.org or at the Gates Medical Research Institute, 1 Kendall Sq., Cambridge, MA 02139.

*A complete list of members of the BCG REVAX Study Team is provided in the Supplementary Appendix, available at NEJM.org.

N Engl J Med 2025;392:1789–800.

DOI: 10.1056/NEJMoa2412381

Copyright © 2025 Massachusetts Medical Society.

CME



A Quick Take
is available at
NEJM.org



APPROXIMATELY 1.7 BILLION PEOPLE ARE estimated to be infected with *Mycobacterium tuberculosis* worldwide, of whom approximately 56 million are newly infected and at high risk for progression to tuberculosis disease.¹ In 2022, an estimated 10.6 million new cases of tuberculosis disease were reported worldwide, which included 1.3 million cases in children younger than 15 years of age.² Tuberculosis disease was responsible for 1.3 million deaths worldwide in 2022, which included 167,000 deaths in persons living with human immunodeficiency virus (HIV) infection.² The incidence of tuberculosis disease in South Africa is among the highest in the world, with an estimated 468 cases per 100,000 persons in 2022.² Cohort studies in the Western Cape province showed that the prevalence of *M. tuberculosis* infection, as assessed by means of an *M. tuberculosis*-specific interferon- γ release assay (IGRA) such as QuantiFERON-TB (QFT) tests, increases quickly among adolescents,^{3,4} with an incidence of QFT test conversion (from negative to positive) as high as 10% per year.³

A previous phase 2 trial assessed the efficacy of the investigational H4:IC31 vaccine and the bacille Calmette–Guérin (BCG) vaccine, as compared with placebo, for protection against *M. tuberculosis* infection in healthy QFT test–negative, HIV-negative participants 12 to 17 years of age in South Africa.³ BCG revaccination was not shown to provide protection against primary *M. tuberculosis* infection, defined by an initial QFT test conversion. However, significant vaccine efficacy was observed for the prevention of sustained *M. tuberculosis* infection, defined by an initial QFT test conversion followed by positive QFT tests at 3 and 6 months thereafter (a secondary end point), with a vaccine efficacy of 45% (95% confidence interval [CI], 6 to 68).³

One interpretation of the significant result for the prevention of sustained *M. tuberculosis* infection is that QFT test reversion (from positive to negative) could indicate successful immune control or clearance of *M. tuberculosis* infection, thus decreasing the risk of progression to disease.⁵ However, because progression to tuberculosis disease occurs in no more than 5 to 10% of persons infected with *M. tuberculosis*, vaccine efficacy for the prevention of sustained *M. tuberculosis* infection may not predict vaccine efficacy for the prevention of disease if the protective immune response leading to the prevention of sustained infection is induced only in the 90 to 95% of

people who would have stayed healthy regardless of vaccination.⁶

In the absence of a licensed vaccine for the prevention of tuberculosis disease in adolescents and adults, we aimed to confirm that BCG revaccination confers protection against sustained *M. tuberculosis* infection in a larger population with an expanded age range and geographic area and to identify candidate correlates of protection for this trial end point. We assumed that a confirmatory trial could potentially support a policy change for BCG revaccination in South Africa or motivate funders to invest in a large phase 3 trial of BCG revaccination for the prevention of disease in support of the global effort to accelerate the end of the tuberculosis disease epidemic.^{7,8} Here we report the results of a phase 2b trial to assess the efficacy of BCG revaccination for the prevention of sustained *M. tuberculosis* infection in South Africa.

METHODS

TRIAL DESIGN AND OBJECTIVES

In this double-blind, phase 2b, randomized, placebo-controlled trial, participants were enrolled at five sites in South Africa: Worcester, Cape Town, and Mbekweni in Western Cape province; Durban in KwaZulu-Natal province; and Johannesburg in Gauteng province. The primary objective was to show the efficacy of BCG revaccination, as compared with placebo, against sustained *M. tuberculosis* infection in QFT test–negative, healthy adolescents. The QFT Gold-Plus test was used for the assessment of *M. tuberculosis* infection.

Secondary objectives were to evaluate the durability of efficacy and the safety and reactogenicity of BCG revaccination. Exploratory objectives were to assess the immunogenicity of BCG revaccination, vaccine efficacy on the basis of alternative definitions of QFT test conversion, and incident cases of laboratory-confirmed tuberculosis disease.

Participants were randomly assigned in a 1:1 ratio to receive the BCG vaccine (BCG Danish 1331 vaccine, AJ Vaccines) or placebo intradermally (Fig. 1). Because the BCG vaccine does not include antigens used in the QFT test, revaccination is not associated with QFT test conversion, and a negative QFT test at day 71 could be used as a criterion for inclusion in the modified intention-to-treat population for the assessment of vaccine efficacy. Randomization was stratified accord-

ing to age, sex, trial site, and school cluster (Worcester site only). The BCG vaccine was administered according to the instructions provided in the product package insert for South Africa (0.1 ml intradermally). Randomization was performed with the use of a validated interactive voice-response or Web-response system. Trial personnel were unaware of the trial-group assignments until the primary end-point analyses were completed. The expected formation of a papule at the BCG vaccine injection site in most vaccine recipients, starting in the second or third week after vaccination, led to the unblinding of the trial-group assignment for most vaccine recipients and site staff at week 2 or 3. This time frame allowed for an unbiased assessment of solicited adverse events from day 1 through day 7.

PARTICIPANTS

Eligible participants were 10 to 18 years of age and HIV-negative and QFT test-negative at screening. Detailed eligibility criteria are described in the Supplementary Appendix and protocol, which are available with the full text of this article at NEJM.org.

END POINTS AND ASSESSMENTS

The primary end point was a sustained QFT test conversion based on positive QFT tests; an assay threshold of 0.35 IU per milliliter of interferon- γ (established by the manufacturer of the QFT test, Qiagen) was used to determine positivity. Sustained *M. tuberculosis* infection was defined by a sustained QFT test conversion from negative to positive, with the initial conversion occurring at any time after a first negative QFT test after randomization, followed by positive QFT tests at 3 and 6 months after conversion. The sampling schedule is shown in Figure 1A.

Secondary end points of the safety and reactivity of BCG revaccination were evaluated on the basis of the adverse events reported, as described in the Supplementary Appendix. Exploratory efficacy end points included an initial QFT test conversion, determined on the basis of various interferon- γ thresholds ranging from 0.36 IU per milliliter to 10 IU per milliliter.

The first 80 participants who were 10 to 12 years of age and had undergone randomization at the Worcester site were enrolled in the immunogenicity subcohort, because this younger age group had not previously been evaluated for immunogenicity. Immunogenicity was assessed

in an exploratory analysis with the use of a whole-blood intracellular cytokine staining assay and was determined from the frequency of BCG vaccine-specific CD4 or CD8 T cells expressing one or more of the following cytokines: interferon- γ , tumor necrosis factor, interleukin-2, interleukin-17, and interleukin-22.

TRIAL OVERSIGHT

An independent data monitoring committee reviewed unblinded safety data every 3 months during the enrollment period and every 6 months thereafter, as well as the outcomes of the primary analyses. The trial was conducted in accordance with the Good Clinical Practice guidelines of the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, regulations of the South African Health Products Regulatory Authority, the institutional review board or independent ethics committee at each site, and other applicable country and local requirements. All the authors were involved in the collection and analysis of the data and the review of the manuscript. The first draft of the manuscript was written by the first author, with assistance from a medical writer funded by the Gates Medical Research Institute. The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol.

STATISTICAL ANALYSIS

The trial was designed to provide 90% power at a one-sided alpha level of 0.025 for the primary end-point analysis. We assumed that for a true vaccine efficacy of 45%, at least 118 sustained QFT test conversions would be needed to show vaccine efficacy with a lower 95% confidence interval boundary of zero. Vaccine efficacy was calculated as 1 minus the hazard ratio for a sustained QFT test conversion (BCG vaccine vs. placebo).

The modified intention-to-treat population included all the participants who had undergone randomization, received the BCG vaccine or placebo, and were QFT test-negative at the day 71 visit or at the first trial visit after day 71 for which a QFT test result was available. The safety population included all the participants who had undergone randomization and received the BCG vaccine or placebo. Results are reported as numbers and percentages for categorical variables and as means with standard deviation or medi-

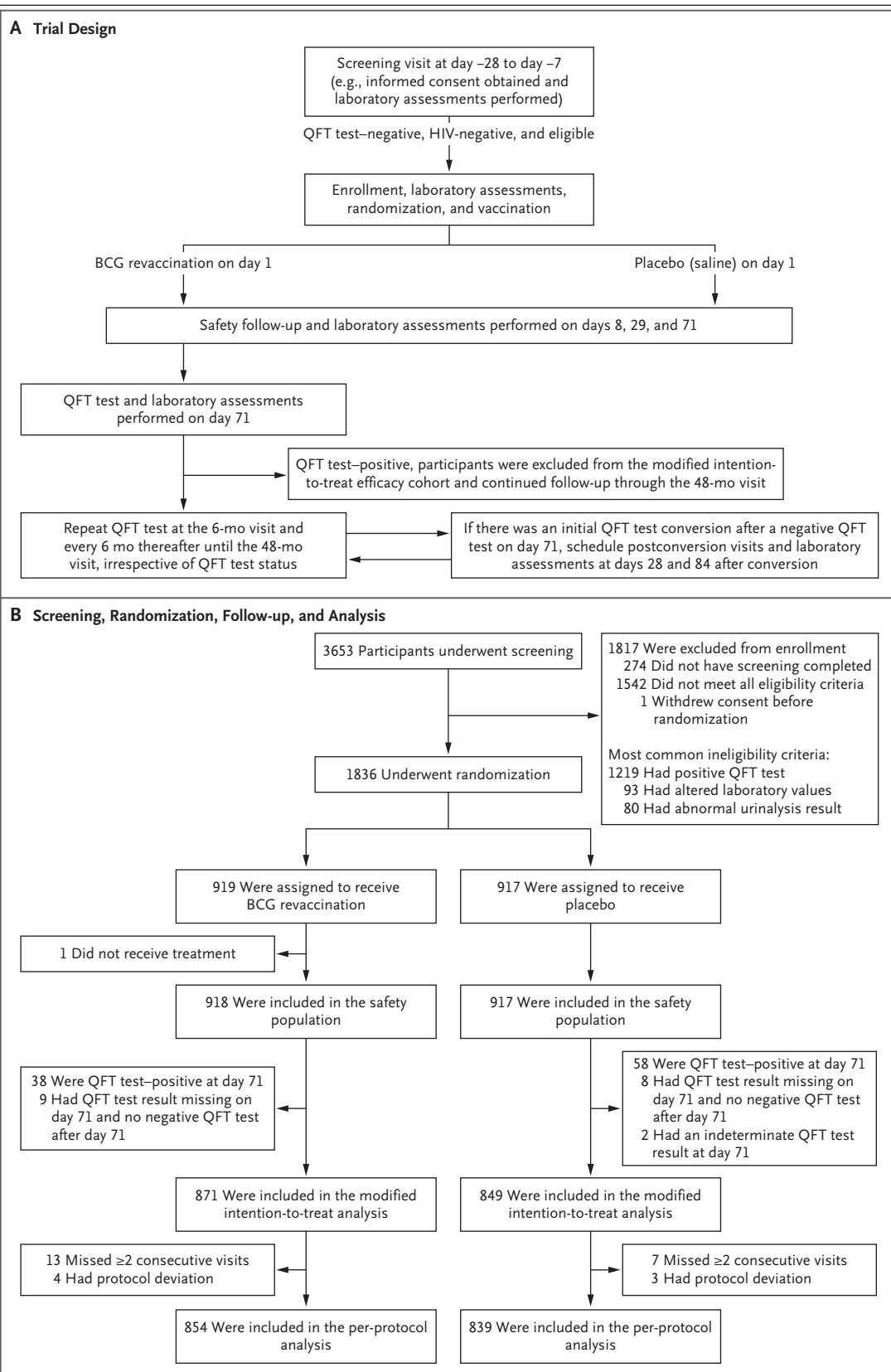


Figure 1 (facing page). Trial Design and Screening, Randomization, Follow-up, and Analyses.

Panel A shows the trial design. Participants were randomly assigned to the bacille Calmette–Guérin (BCG)–vaccine group or the placebo group, and QuantiFERON-TB (QFT) tests were performed according to trial schedule. Participants who had a QuantiFERON-TB (QFT) test conversion (negative to positive) at day 71 (or if the visit was missed or not feasible to conduct, at the next feasible visit) were excluded from the primary modified intention-to-treat efficacy analysis but were followed for safety and efficacy to the end of the trial. Each participant was followed for safety for a minimum of 6 months after vaccination. Panel B shows the screening, randomization, follow-up, and analysis. Among 3653 persons who had undergone screening, 1817 were excluded from enrollment. The most common ineligibility criteria were a positive QFT test, altered laboratory values, and an abnormal urinalysis result. Protocol deviations leading to exclusion from the modified intention-to-treat population included nonadherence to the investigational product (3 participants in the BCG-vaccine group) and not meeting all eligibility criteria (1 participant in the BCG-vaccine group and 3 in the placebo group). A total of 1836 underwent randomization in a 1:1 ratio to receive BCG revaccination (919 participants) or placebo (917 participants).

ans with minimum–maximum ranges for continuous variables.

The primary efficacy end point was analyzed with the use of a log-rank test, stratified according to sex and age group (10 or 11 years, 12 to 14 years, and >14 years) to evaluate differences in distributions of event times between the BCG-vaccine group and placebo group. Hazard ratios and 95% confidence intervals were estimated from a stratified Cox proportional-hazards model. Adverse events in different categories were summarized according to trial group, and the 95% confidence intervals were calculated for each proportion of participants with the use of the conditional binomial Clopper–Pearson method with mid-P correction. Adverse events were coded according to the *Medical Dictionary for Regulatory Activities*, version 26.0 or higher. Adverse events were graded according to the Division of AIDS Table for Grading the Severity of Adult and Pediatric Adverse Events, version 2.1.⁹

The percentage of missing data was low, with an incidence of early discontinuation of 5.1% in the BCG-vaccine group and 3.9% in the placebo group, which included 1.0% and 0.2% of participants, respectively, who were lost to follow-up and 2.5% of participants in each group who

withdrew from the trial. Given that the missingness was minimal and did not appear to be dependent on known factors, we assumed that the data were missing completely at random, and therefore only the observed data were used in analyses.

RESULTS

BASELINE CHARACTERISTICS

From October 16, 2019, to July 22, 2021, a total of 3653 participants underwent screening, and 1836 were enrolled and underwent randomization (Fig. 1B). The representativeness of the trial participants is shown in Table S1 in the Supplementary Appendix. At the time of the primary analysis (data cutoff date, June 30, 2023), a total of 1752 participants had remained in the trial, and 83 (4.5%) had discontinued the trial early. Among the 3653 participants who had undergone screening, 307 had no blood drawn, 12 had indeterminate QFT test results, and 1 had invalid results, and of the remaining 3333 participants with screening QFT test results available, 1219 (36.6%) were QFT test–positive (Table S2). The percentage of participants with positive QFT tests at screening was higher at sites in the Western Cape (37.8 to 39.6%) than in KwaZulu-Natal (29.1%) or Gauteng (23.6%) provinces. Demographic and clinical characteristics of the participants at baseline were similar in the two groups (Table S3). The median age of the participants was 13 years (range, 10 to 18), and most participants (1464 [79.8%]) identified as Black African.

EFFICACY

The modified intention-to-treat population included 871 participants in the BCG-vaccine group and 849 participants in the placebo group (Fig. 1B). Among these participants, 38 (4.1%) in the BCG-vaccine group and 58 (6.3%) in the placebo group were excluded from the modified intention-to-treat population because they had a positive QFT test at day 71; 9 (1.0%) and 8 (0.9%) participants, respectively, were excluded because test results at day 71 were missing and no negative QFT test was documented at the first visit after day 71; and 2 participants (0.2%) in the placebo group were excluded because of indeterminate results at day 71. In the primary analysis, which was performed after a median of 30 months (inter-

Table 1. Sustained and Initial QFT Test Conversions.*

Variable	BCG Vaccine (N=871)	Placebo (N=849)
Initial QFT test conversion on or after day 71 — no. (%)†	135 (15.5)	125 (14.7)
Not evaluable for QFT test reversion because of missing or indeterminate QFT results — no.	24	24
QFT test reversion at day 84 or 6 mo after conversion — no.	49	42
Sustained QFT test conversion‡		
No. with event	62	59
Proportion (95% CI)§	0.0712 (0.0541 to 0.0883)	0.0695 (0.0524 to 0.0866)
Person-time accrued — mo¶	19,719	19,383
Overall incidence — no. of events per person-mo (95% CI)	0.0031 (0.0024 to 0.0040)	0.0030 (0.0023 to 0.0039)
Hazard ratio (95% CI)**	1.04 (0.73 to 1.48) ††	—
Vaccine efficacy — % (95% CI) ‡‡	−3.8 (−48.3 to 27.4)	—
Initial QFT test conversion — proportion (95% CI)§	0.16 (0.13 to 0.18)	0.15 (0.12 to 0.17)
Person-time accrued — mo¶	19,695	19,371
Overall incidence — no. of events per person-mo (95% CI)	0.0069 (0.0057 to 0.0081)	0.0065 (0.0054 to 0.0077)

* All analyses were performed in the modified intention-to-treat population, which included all the participants who had undergone randomization, received the bacille Calmette–Guérin (BCG) vaccine or placebo, and were QuantiFERON-TB (QFT) test–negative at the day 71 visit or at the first trial visit after day 71 for which a QFT test result was available. A QFT test conversion was defined as a change from negative (<0.35 IU per milliliter) on day 71 (or the first visit after a missed day 71 visit) to positive (≥0.35 IU per milliliter). A QFT test reversion was defined as a change from positive to negative. The QFT Gold-Plus test was used for the assessments for *M. tuberculosis* infection.

† Participants with a first QFT test conversion on or after day 71 (if the day 71 visit was missed) were counted for this end point, irrespective of whether a change from positive to negative occurred on day 84 or at the 6-month visit.

‡ Participants who had a primary QFT test conversion and continued to have a positive test at both 3 and 6 months after the initial conversion were counted for this end point.

§ Proportion was calculated as the number of participants with a QFT test conversion divided by the total number of participants. The 95% confidence intervals were calculated with the use of the Miettinen and Nurminen method without stratification.¹⁶

¶ Person-time was calculated as the time (in months) to conversion if observed or as the censored time if conversion was not observed.

|| Exact confidence intervals were based on a Poisson process with constant intensity.

** The hazard ratio was calculated with the use of a stratified Cox proportional-hazards model with sex and age group (10 or 11 years, 12 to 14 years, and >14 years) as stratification variables.

†† One-sided $P=0.58$ by log-rank test, stratified according to sex and age group.

‡‡ Vaccine efficacy was calculated as 1 minus the hazard ratio (BCG vaccine vs. placebo) for a sustained QFT test conversion. The lower limit of the 95% confidence interval for vaccine efficacy was calculated as 1 minus the upper limit of the 95% confidence interval for the hazard ratio, and the upper limit of the 95% confidence interval for vaccine efficacy as 1 minus the lower limit of the hazard ratio.

quartile range, 25 to 33) of follow-up, a sustained QFT test conversion was observed in 62 participants (7.1%; 95% CI, 5.4 to 8.8) in the BCG-vaccine group and 59 participants (7.0%; 95% CI, 5.2 to 8.7) in the placebo group (one-sided $P=0.58$) (Table 1 and Fig. 2A and 2B). The hazard ratio for a sustained test conversion (BCG vaccine vs. placebo) was 1.04 (95% CI, 0.73 to 1.48), for a vaccine efficacy point estimate of −3.8% (95% CI, −48.3 to 27.4). Both the percentage of participants with a sustained QFT test

conversion and the cumulative incidence of a sustained QFT test conversion over time were similar in the trial groups. (Table 1). Site-specific point estimates of vaccine efficacy are listed in Table S4.

Overall, after day 71, an initial QFT test conversion was observed in 135 participants (15.5%) in the BCG-vaccine group and 125 participants (14.7%) in the placebo group (Table 1 and Fig. 2C). The incidence of QFT test reversion was similar in the trial groups (Table 1 and Table S5). Among

the participants who had an initial QFT test conversion and three consecutive positive or negative QFT tests available, 62 of 111 participants (56%) in the BCG-vaccine group and 59 of 101 participants (58%) in the placebo group had a sustained QFT test conversion, whereas 49 (44%) and 42 (42%), respectively, had a QFT test reversion at day 84 or 6 months after conversion.

The results for an initial QFT test conversion that were based on a range of increasingly stringent thresholds were similar in the trial groups, regardless of the threshold (Fig. 2D and Table S6). For example, an initial QFT test conversion based on a QFT threshold of 4.0 IU per milliliter was observed in 60 participants (6.9%) in the BCG-vaccine group and 64 participants (7.5%) in the placebo group (Fig. 2D, Table S6, and Fig. S1). Laboratory-confirmed tuberculosis disease developed in 6 participants in the safety population (3 in the BCG-vaccine group and 3 in the placebo group).

SAFETY

Solicited local adverse events were reported by 77.8% participants in the BCG-vaccine group and 38.1% in the placebo group; 40.5% and 34.3%, respectively, reported having any solicited systemic adverse event (Table 2). Swelling was the most common solicited local adverse event, and tiredness was the most common solicited systemic adverse event.

Unsolicited nonserious adverse events occurred in 185 participants (20.2%) in the BCG-vaccine group and 117 participants (12.8%) in the placebo group (Table 2). The most common unsolicited nonserious adverse events in the BCG-vaccine group were headache (in 3.9%) and injection-site pain (in 2.3%). Most adverse events were mild to moderate in severity. Three participants (0.3%) in each group had serious adverse events, which were determined by the site investigator not to be related to the trial regimen. No serious adverse events leading to death, serious adverse drug reactions, or adverse events leading to premature trial discontinuation were reported. In the BCG-vaccine group, 32 participants had an injection-site scar greater than 10 mm in its largest diameter. Among these participants, 9 had injection-site scars larger than 15 mm, and 2 had scars larger than 20 mm. One large keloid (2.5 cm in its largest diameter) was surgically removed at the participant's request.

IMMUNOGENICITY

The frequencies of antigen-specific CD4 T cells expressing any combination of interferon- γ , tumor necrosis factor, interleukin-2, interleukin-17, or interleukin-22 were higher with BCG revaccination than with placebo, which remained higher than baseline 6 months after vaccination (Fig. 3 and Figs. S2 and S3). Type 1 helper T-cell responses (CD4 T cells expressing interleukin-2, interferon- γ , or tumor necrosis factor) predominantly accounted for this observation, rather than CD4 T cells expressing interleukin-17 or interleukin-22. A transient increase of CD8 responses to the BCG vaccine was observed at day 29 after vaccination (Fig. S4).

DISCUSSION

This trial was conducted to assess the findings of an earlier trial involving QFT test-negative, HIV-negative adolescents that showed a significantly lower incidence of sustained QFT test conversion with BCG revaccination than with placebo.³ We assumed that confirming the protective effect of BCG revaccination against sustained *M. tuberculosis* infection, along with providing correlates of protection data, could motivate funders to support a future trial evaluating disease prevention. Although a vaccine efficacy point estimate of 45.4% (95% CI, 6.4 to 68.1) with respect to the prevention of sustained *M. tuberculosis* infection was observed in the earlier trial, efficacy was not observed in the current trial (vaccine efficacy point estimate, -3.8%; 95% CI, -48.3 to 27.4). The differences in observed efficacy are unlikely due to differences in BCG strain, case definition, biospecimen processing, or assay performance, which were intentionally harmonized. Our trial differed from the previous trial in that it was powered to assess the efficacy of BCG revaccination in the prevention of sustained QFT test conversion (the primary end point) with the use of the QFT Gold-Plus test rather than the QFT Gold In-Tube test. In addition, the number of participants in each trial group in our trial was approximately three times as large as that in the previous trial, and we broadened the age range to include children 10 and 11 years of age, as well as the geographic area to include additional communities in the Western Cape, Gauteng, and KwaZulu-Natal provinces, which have a more diverse population.

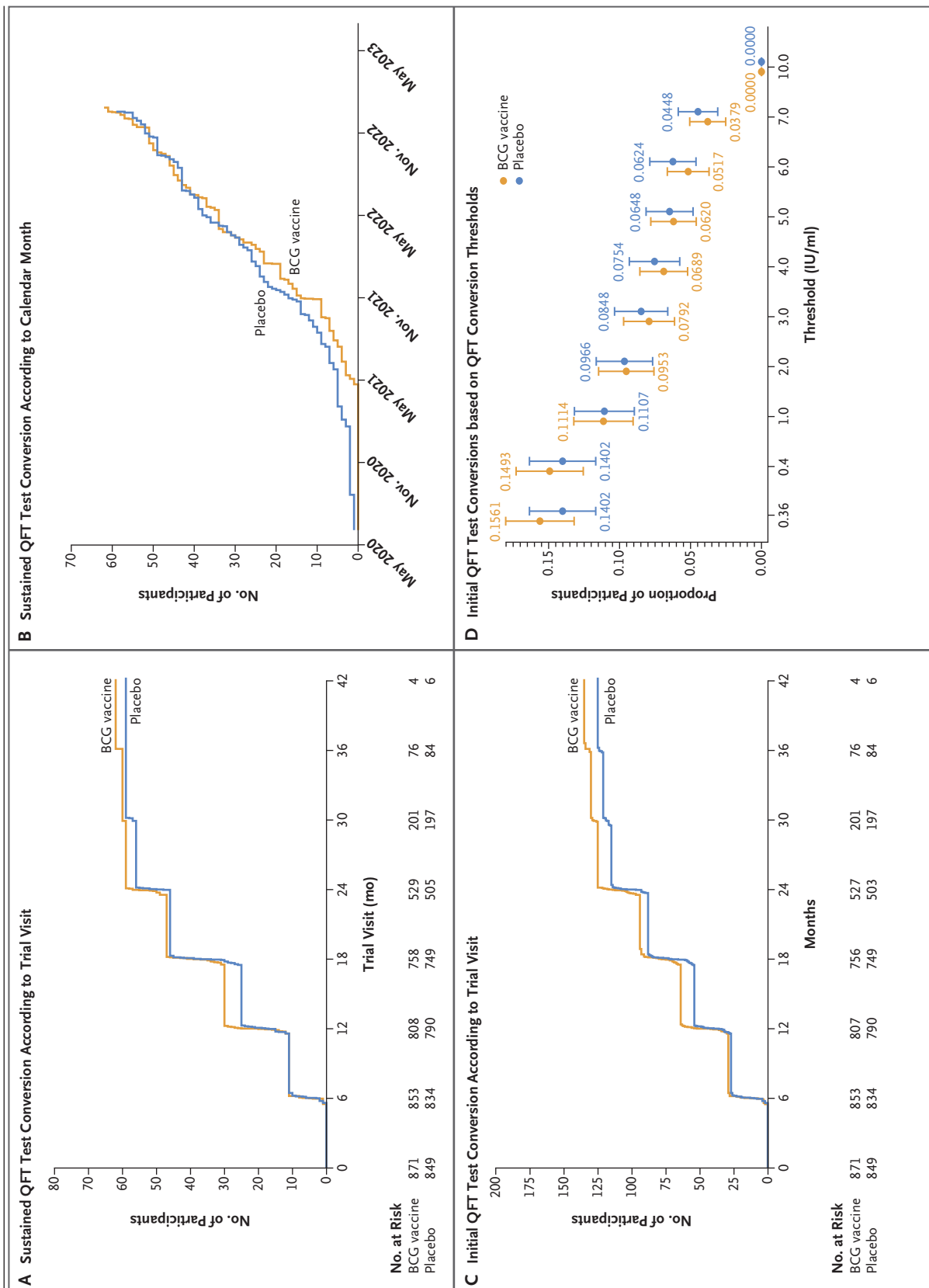


Figure 2 (facing page). Vaccine Efficacy in the Modified Intention-to-Treat Population.

Panel A shows the cumulative event curves for a sustained QFT test conversion (primary efficacy end point) according to trial visit. Panel B shows the cumulative event curves for a sustained QFT test conversion according to calendar month. Panel C shows the cumulative event curves for an initial QFT test conversion according to trial visit. Panel D shows the proportion of participants with an initial QFT test conversion based on a range of increasingly stringent QFT test conversion thresholds higher than 0.35 IU per milliliter. I bars represent 95% confidence intervals.

However, geographic or ethnic differences between the trials are unlikely to explain the observed differences in vaccine efficacy, since the observed site-specific vaccine efficacy in this trial at the Worcester site, where 92.7% of the participants in the earlier trial were enrolled, was similar to the overall vaccine efficacy. The overall incidence of initial QFT test conversion in this trial (6.5 to 6.9% per person-year) was lower than the overall incidence reported in the earlier trial (9.9% per person-year). This finding could be due in part to the lockdown related to the coronavirus disease 2019 (Covid-19) pandemic, with extended school closures in 2020 and 2021,¹⁰ and inclusion of trial sites with a lower force of infection (the rate at which susceptible persons acquire an infectious disease). For example, two sites outside the Western Cape province enrolled 20% of participants but contributed only 5.0% [6 of 121] of the participants with a primary end-point event. One observation that differed between this trial and the earlier trial — and which may have contributed to the difference in the observed vaccine efficacy — is that the incidence of QFT test reversion in the placebo group was higher in our trial (42%) than in the earlier trial (25%), whereas the incidence of QFT test reversion in the BCG-vaccine group was similar (44% vs. 46%). Whether the differences in baseline characteristics (with the inclusion of younger children and a more diverse population in the current trial) or the lower intensity of *M. tuberculosis* exposure (such as that related to Covid-19 lock-down restrictions) contributed to the observed difference in the incidence of reversion remains unknown.

The safety and reactogenicity data reported here correspond with the known safety profile of the marketed BCG vaccine. This vaccine is well

characterized. Larger scars (>10 mm) at the administration site of BCG vaccine were observed in approximately 3% of the BCG vaccine recipients, a finding consistent with previous reports.¹¹

A recent trial of BCG revaccination for the prevention of sustained *M. tuberculosis* infection in 2000 adult health care workers in Brazil showed a lack of vaccine efficacy against either initial or sustained QFT test conversion.¹² Given that a lack of vaccine efficacy was observed in two randomized, controlled trials that involved participants with different ancestry and age range and were conducted in locations with a different force of infection, geography, climate, and environmental mycobacteria exposure, the evidence supporting BCG revaccination for the prevention of *M. tuberculosis* infection appears to have weakened. However, our data do not allow conclusions to be drawn on the potential efficacy of BCG revaccination for the prevention of disease, for which supporting evidence is limited.^{13,14} A trial evaluating prevention of disease with BCG revaccination trial is currently under way in India and may provide informative data on that end point.¹⁵

Considerable attention has been given to the predictive power of the QFT test for tuberculosis disease, and higher interferon- γ conversion values were previously reported to be associated with a higher risk of progression to tuberculosis disease.^{13,14} However, for the prevention of infection, we observed no changes in vaccine efficacy with increasing interferon- γ thresholds, a finding that argues against the possibility that participants with false-positive conversions contributed to the observed lack of vaccine efficacy.

The strengths of this trial include the use of stringent statistical criteria to assess vaccine efficacy, and the unambiguous results for the primary end point. BCG revaccination did not provide any protection from *M. tuberculosis* infection as assessed with the use of the QFT test, regardless of whether conversion to infection was defined by a sustained conversion, initial conversion, or conversion with higher interferon- γ thresholds. The trial groups were well balanced in terms of baseline characteristics, and the trial was conducted with stringent monitoring and oversight. The level of participant retention and community engagement was high throughout the trial. A limitation of the trial was that enrollment was paused for 4 months because of restrictions related to the Covid-19 pandemic,

Table 2. Incidence of Adverse Events and Medication Use.*

Adverse Event or Medication Use	BCG Vaccine (N=918)	Placebo (N=917)
Solicited adverse events and medication use		
Any injection-site symptom		
No. with event/total no.	754/915	381/915
Percent (95% CI)	82.4 (79.8 to 84.8)	41.6 (38.5 to 44.9)
Any general body symptom		
No. with event/total no.	371/916	314/915
Percent (95% CI)	40.5 (37.4 to 43.7)	34.3 (31.3 to 37.4)
Use of medication 1 to 7 days after vaccination		
No. with event/total no.	30/903	27/899
Percent (95% CI)	3.3 (2.3 to 4.6)	3.0 (2.0 to 4.3)
Injection-site symptoms according to severity†		
Swelling — no./total no. (%)		
Any severity	657/912 (72.0)	248/910 (27.3)
Severe or life-threatening	2/912 (0.2)	1/910 (0.1)
Pain — no./total no. (%)		
Any severity	381/914 (41.7)	148/915 (16.2)
Severe or life-threatening	17/914 (1.9)	6/915 (0.7)
Redness — no./total no. (%)		
Any severity	349/912 (38.3)	111/911 (12.2)
Severe or life-threatening	0	0
Solicited systemic adverse events according to severity†		
Tiredness — no./total no. (%)		
Any severity	229/913 (25.1)	184/913 (20.2)
Severe or life-threatening	8/913 (0.9)	6/913 (0.7)
Headache — no./total no. (%)		
Any severity	195/913 (21.4)	164/913 (18.0)
Severe or life-threatening	7/913 (0.8)	9/913 (1.0)
Stomach problems — no./total no. (%)		
Any severity	136/913 (14.9)	140/913 (15.3)
Severe or life-threatening	8/913 (0.9)	8/913 (0.9)
Fever — no./total no. (%)‡		
Any severity	24/911 (2.6)	27/907 (3.0)
Severe or life-threatening	5/911 (0.5)	5/907 (0.6)
Adverse-event categories§		
Mild adverse events		
Any mild event — no.	315	90
Percent (95% CI)	34.3 (31.3 to 37.4)	9.8 (8.0 to 11.9)
Moderate adverse events		
Any moderate event — no.	47	28
Percent (95% CI)	5.1 (3.8 to 6.7)	3.1 (2.1 to 4.3)
Severe adverse events		

Table 2. (Continued.)

Adverse Event or Medication Use	BCG Vaccine (N=918)	Placebo (N=917)
Any severe event — no.	5	3
Percent (95% CI)	0.5 (0.2 to 1.2)	0.3 (0.1 to 0.9)
Related adverse events¶		
Any related event — no.	273	9
Percent (95% CI)	29.7 (26.8 to 32.8)	1.0 (0.5 to 1.8)
Related severe adverse events¶		
Any related severe event — no.	1	0
Percent (95% CI)	0.1 (0 to 0.5)	—
Unsolicited nonserious adverse events		
Any unsolicited nonserious event — no.	185	117
Percent (95% CI)	20.2 (17.7 to 22.8)	12.8 (10.7 to 15.0)
Serious adverse events		
Any serious event — no.	3	3
Percent (95% CI)	0.3 (0.1 to 0.9)	0.3 (0.1 to 0.9)
Serious adverse event with outcome of death — no.	0	0
Serious adverse drug reaction — no.¶	0	0
Adverse events leading to premature trial discontinuation — no.	0	0
Adverse events of special interest		
Any event of special interest — no.	5	0
Percent (95% CI)	0.5 (0.2 to 1.2)	—
Most common nonserious adverse events**		
Headache — no. (%)	36 (3.9)	27 (2.9)
Injection-site pain — no. (%)	21 (2.3)	0
Injection-site ulceration — no. (%)	20 (2.2)	0
Pyrexia — no. (%)	12 (1.3)	3 (0.3)
Oropharyngeal pain — no. (%)	9 (1.0)	4 (0.4)
Upper respiratory tract infection — no. (%)	8 (0.9)	15 (1.6)

* All analyses were performed in the safety population, which included all the participants who had undergone randomization and received the BCG vaccine or placebo. Adverse events were coded according to the *Medical Dictionary for Regulatory Activities*, version 26.0. Data on solicited adverse events are based on the adverse events reported by the participant on a diary card from days 1 to 7. Data on unsolicited adverse events were collected from the time of screening through day 28. Data on serious adverse events were collected from the time of screening through month 6. Data on serious adverse drug reactions were collected from day 1 through the end of trial. A participant with multiple adverse events within a system organ class or preferred term was counted only once. The 95% confidence intervals were calculated with the use of the conditional binomial Clopper–Pearson method with mid-P correction.

† Participants were categorized according to the highest severity experienced.

‡ Fever was defined as a body temperature of 38.0°C or higher.

§ An adverse event with changing severity was counted only once according to the highest severity.

¶ Related events were those determined by the site investigator to be related to the assigned regimen.

|| A serious adverse drug reaction was defined as a serious adverse event that was determined by a site investigator to be related to trial regimen.

** The most common unsolicited adverse events were those that occurred in at least 1% of the participants in any group.

and schools were closed for several additional months,¹⁰ which may have contributed to a lower incidence of QFT test conversions.

Although this trial does not allow us to draw firm conclusions on the efficacy of BCG revaccination for the prevention of disease, the lack

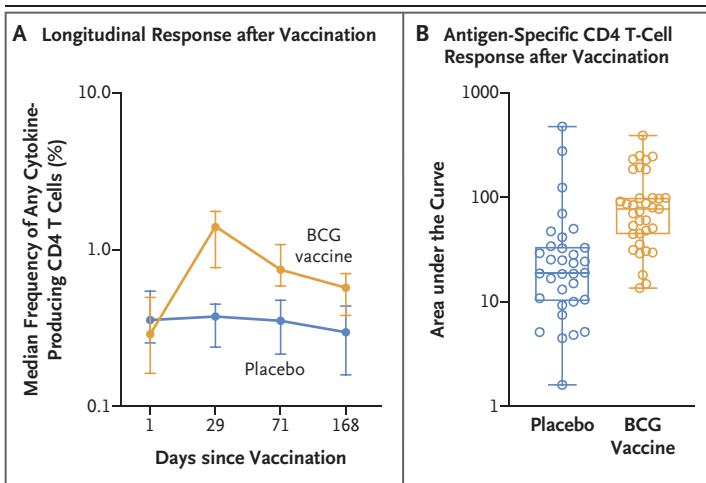


Figure 3. Immunogenicity of BCG Revaccination.

Shown are the frequencies of antigen-specific CD4 T cells expressing any combination of interferon- γ , tumor necrosis factor, interleukin-2, interleukin-17, or interleukin-22 after stimulation with BCG vaccine, as measured with the use of whole-blood intracellular cytokine staining assay in participants who received BCG revaccination (orange) or placebo (blue). Panel A shows longitudinal responses (plotted on a logarithmic scale) during the first 168 days after vaccination. I bars indicated 95% confidence intervals. Panel B shows the area under the curve (plotted on a logarithmic scale) for BCG-specific CD4 T-cell frequencies per participant during the first 168 days after vaccination. Each circle represents an individual participant. The horizontal line within each box indicates the median, the box indicates the interquartile range (quartile 1 to quartile 3), and the whiskers indicate the minimum and maximum values.

BCG revaccination conferring protection against disease.

Supported by the Gates Medical Research Institute through grants from the Gates Foundation (INV-008522 and INV-057216); the Gates Medical Research Institute is a wholly owned subsidiary of the Gates Foundation.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

We thank the trial participants and their families for taking part in this trial; the members of the community advisory board and independent data monitoring committee; the staff at the trial sites and at IQVIA, MMS Holdings, and Bio Analytical Research Corporation South Africa for their contributions; and Madeeha Aqil, Ph.D., for medical writing assistance with an earlier version of the manuscript, which was performed in accordance with Good Publication Practice guidelines.

AUTHOR INFORMATION

Alexander C. Schmidt, M.D.,¹ Lee Fairlie, M.B., Ch.B.,² Elizabeth Hellström, M.B., Ch.B.,³ Angelique Luabeya Kany Kany, M.B., Ch.B.,⁴ Keren Middelkoop, M.B., Ch.B.,⁵ Kogieleum Naidoo, Ph.D.,⁶ Gonasagrie Nair, M.B., Ch.B.,⁵ Anele Gela, Ph.D.,⁴ Elisa Nemes, Ph.D.,⁴ Thomas J. Scriba, Ph.D.,⁴ Amy Cinar, Ph.D.,¹ Nicole Frahm, Ph.D.,¹ Robin Mogg, Ph.D.,¹ David Kaufman, M.D.,¹ Michael W. Dunne, M.D.,¹ and Mark Hatherill, M.D.⁴

¹Gates Medical Research Institute, Cambridge, MA; ²Wits Reproductive Health and HIV Institute, University of Witwatersrand, Johannesburg; ³Be Part Yoluntu Centre, Paarl, South Africa; ⁴South African Tuberculosis Vaccine Initiative, Institute of Infectious Disease and Molecular Medicine and Department of Pathology, University of Cape Town, Cape Town, South Africa; ⁵Desmond Tutu HIV Centre, Institute of Infectious Disease and Molecular Medicine, University of Cape Town, Cape Town, South Africa; ⁶South African Medical Research Council HIV-TB Pathogenesis and Treatment Research Unit, Centre for the AIDS Programme of Research in South Africa, Durban.

of vaccine efficacy with respect to prevention of infection probably decreases the likelihood of

REFERENCES

- Houben RMGJ, Dodd PJ. The global burden of latent tuberculosis infection: a re-estimation using mathematical modelling. *PLoS Med* 2016;13(10):e1002152.
- Global tuberculosis report 2023. Geneva: World Health Organization, November 17, 2023 (<https://www.who.int/publications/i/item/9789240083851>).
- Nemes E, Geldenhuys H, Rozot V, et al. Prevention of *M. tuberculosis* infection with H4:IC31 vaccine or BCG revaccination. *N Engl J Med* 2018;379:138-49.
- Mahomed H, Hawkrigde T, Verver S, et al. The tuberculin skin test versus QuantiFERON TB Gold in predicting tuberculosis disease in an adolescent cohort study in South Africa. *PLoS One* 2011;6(3):e17984.
- Dale KD, Schwalb A, Coussens AK, et al. Overlooked, dismissed, and downplayed: reversion of *Mycobacterium tuberculosis* immunoreactivity. *Eur Respir Rev* 2024;33:240007.
- Nemes E, Fiore-Gartland A, Boggiano C, et al. The quest for vaccine-induced immune correlates of protection against tuberculosis. *Vaccine Insights* 2022;1:165-81.
- Sumner T, Clark RA, Mukandavire C, et al. Modelling the health and economic impacts of M72/AS01_E vaccination and BCG revaccination: estimates for South Africa. *Vaccine* 2024;42:1311-8.
- Clark RA, Sumner T, Weerasuriya CK, Bakker R, Scriba TJ, White RG. Estimating the potential public health value of BCG revaccination. *J Infect Dis* 2024;230(1):e139-e143.
- Division of AIDS. Division of AIDS (DAIDS) table for grading the severity of adult and pediatric adverse events, corrected version 2.1. Bethesda, MD: National Institute of Allergy and Infectious Diseases, July 2017 (<https://rsc.niaid.nih.gov/sites/default/files/daidsgradingcorrectedv21.pdf>).
- Shepherd D, Mohohlwane N. The impact of COVID-19 in education — more than a year of disruption. Wave 5. Johannesburg: National Income Dynamics (NIDS)-Coronavirus Rapid Mobile Survey (CRAM), July 8, 2021 (<https://crams.survey.org/wp-content/uploads/2021/07/11.-Shepherd-D.-.-Mohohlwane-N.-2021.-Changes-in-education-A-reflection-on-COVID-19-effects-over-a-year.pdf>).
- Fang JW, Ko BM, Wilson JA. BCG vaccination scars: incidence and acceptance amongst British high-school children. *Child Care Health Dev* 1993;19:37-43.
- Dos Santos PCR, Messina NL, de Oliveira RD, et al. Effect of BCG vaccination against *Mycobacterium tuberculosis* infection in adult Brazilian health-care workers: a nested clinical trial. *Lancet Infect Dis* 2024;24:594-601.
- Barreto ML, Pereira SM, Pilger D, et al. Evidence of an effect of BCG revaccination on incidence of tuberculosis in school-aged children in Brazil: second report of the BCG-REVAC cluster-randomised trial. *Vaccine* 2011;29:4875-7.
- Glynn JR, Fielding K, Mzembe T, et al. BCG re-vaccination in Malawi: 30-year follow-up of a large, randomised, double-blind, placebo-controlled trial. *Lancet Glob Health* 2021;9(10):e1451-e1459.
- Prasad R. BCG revaccination study in high-risk adults to begin in 23 states. December 2, 2023. *The Hindu*. December 3, 2023 (<https://www.thehindu.com/sci-tech/bcg-revaccination-study-in-high-risk-adults-to-begin-in-23-states/article67594580.ece>).
- Miettinen O, Nurminen M. Comparative analysis of two rates. *Stat Med* 1985;4:213-26.

Copyright © 2025 Massachusetts Medical Society.